Dr. Sushovan Majhi

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Research Interests

Applied Topology, **Topological Data Analysis (TDA)**, **Discrete and Computational Geometry**, **Mathematical Foundations of Machine Learning**, **Certified and Interpretable Statistical Learning**, and **Reconstruction and Comparison of Shapes and Graphs**.

One question organizes the work: *inference for data where the manifold hypothesis fails*—at branch points, corners, intersections, and vanishing reach—with certificates, and with proofs of where certificates cannot exist. It is pursued in three movements. First, theorems about when a finite sample remembers the shape it was drawn from. Second, guarantees for what happens next: what a topological descriptor is forbidden to lose once it is handed to a classifier. Third, the collaborations—in climate, fluid mechanics, geophysics, and ecology—where the theorems get tested on data that is unrepentantly real.

Education

- **Doctor of Philosophy** in **Mathematics** **August 2014–December 2020**
  Tulane University, New Orleans, LA, USA.
  Advisor: **Prof. Carola Wenk**
  Courses: computational geometry, computational topology, topological data analysis, differential geometry, differentiable manifolds, algorithms, data structures, computational complexity, and scientific computing.

- **Master of Science** in **Mathematics** **August 2009–May 2012**
  Tata Institute of Fundamental Research, Bangalore, India
  Courses: ordinary and partial differential equations, probability theory, complex analysis, functional analysis, numerical linear algebra, measure theory, mechanics.

- **Bachelor of Science** in **Mathematics** (Hons.) **July 2006–May 2009**
  Ramakrishna Mission Vidyamandira, **Calcutta University**, West Bengal, India
  Courses: calculus, real analysis, linear algebra, numerical analysis, game theory, statistics, physics.

Work Experience

- **Assistant Professor** **August 2024–*current***
  Data Science Program, George Washington University, Washington D.C., USA

- **Visiting Assistant Professor** **August 2023–July 2024**
  Data Science Program, George Washington University, Washington D.C., USA

- **Postdoctoral Research Fellow** **January 2021–July 2023**
  School of Information, University of California, Berkeley, USA
  Role: The responsibilities included conducting research broadly in data science, forging new research collaborations, organizing research webinar series.

Editorial, Program & Professional Leadership

- **Program Committee**, **Theory Track** (Track T), 43rd International Symposium on Computational Geometry (SoCG 2027), part of CG Week 2027. Bangalore, India, June 2027.

- Founding Series Editor (with Atish Mitra), *Applied Geometry & Topology*, Chapman & Hall/CRC, 2026–present.

- Invited by the American Mathematical Society to propose an edited volume in the *Contemporary Mathematics* series, arising from the AMS Special Session on Artificial Intelligence and Machine Learning Methods in Topological Data Analysis (Washington, DC, October 2026).
- Invited contributor, AMS *Contemporary Mathematics* volume on Geometric Data Science, guest-edited by Egon Schulte, Olga Anosova and Vitaliy Kurlin. Invited July 2026.
- **Organizer**, AMS Special Session on "Artificial Intelligence and Machine Learning Methods in Topological Data Analysis", AMS Eastern Sectional Meeting, George Washington University, October 3–4, 2026.

PREPRINTS

Authors are listed alphabetically in mathematics and theoretical computer science; papers in machine learning and the applied sciences list authors by contribution.

15. Buddha Nath Sharma, Joe Opitz*, Adam Moser*, Sayam Palrecha, **SM**, Atish Mitra. Mixup Barcodes for Topology-Aware Financial Decision Making, 22 pages, October 2026. Available at: arXiv:2610.12396 [stat.AP]

14. Pramita Bagchi, Edward Bae*, Atish Mitra, Alexander D. Silberman, Žiga Virk, **SM**. Where Do Two Populations of Persistence Diagrams Differ? Calibrated Local Inference at a Fixed Budget, 45 pages, October 2026. Available at: arXiv:2610.08292 [stat.ME]
    • Under review

13. **SM**, Pramita Bagchi. The Null Is the Hard Part: Exact Tests for Memorization in Generative Models, 23 pages, October 2026. Available at: arXiv:2610.00251 [cs.LG]
    • Under review

12. **SM**, Atish Mitra, B. Todd Hoffman, David Rathgeber, Burt Todd. Tracking-Free Topological Measurement in Vertical Gas–Liquid Pipe Flow: Calibration, Precision, and Selectivity, October 2026. Available at: doi:10.5281/zenodo.23082912

11. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. COMPLEX: A Closed-Form Certified Embedding of Multiparameter Persistence Modules, 40 pages, September 2026. Available at: arXiv:2609.22012 [cs.LG]
    • Under review

10. **SM**. Certified Topological Interaction in Neural Representations: Exact Tests and the Statistic They Require, 36 pages, September 2026. Available at: arXiv:2609.08561 [cs.LG]
    • Under review

9. Pramita Bagchi, **SM**, Atish Mitra, Žiga Virk. Statistical Inference for Persistence Diagrams via Landmark Embeddings: Minimax Theory and Finite Approximation, 87 pages, September 2026. Available at: arXiv:2609.07691 [math.ST]
    • Under review at *Electronic Journal of Statistics* (EJS)

8. Kazuhiro Kawamura, **SM**, Atish Mitra. The Intersection Euler Characteristic Profile: Euler Calculus and Stability for Topological Interaction of Ball Unions, August 2026. Available at: arXiv:2608.06180 [math.AT]
    • Under review at *Journal of Applied and Computational Topology* (JACT)

7. **SM**. A constant-factor approximation of the Gromov–Hausdorff distance in the plane, June 2026. Available at: arXiv:2606.17051 [cs.CG]

6. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. A Closed-Form Adaptive-Landmark Kernel for Certified Point-Cloud and Graph Classification, May 2026. Available at: arXiv:2605.04046 [cs.LG]
    • Under review at *Journal of Machine Learning Research* (JMLR)

* Undergraduate student at the time of research or submission

5. **SM**, Atish Mitra, Santanu Nandi, Md. Nurujjaman, Buddha Nath Sharma. Detecting Regime Transitions in Dynamical Systems via the Mixup Euler Characteristic Profile, April 2026. Available at: arXiv:2604.15262 [math.DS]
   • Under review at *Chaos: An Interdisciplinary Journal of Nonlinear Science*

4. Brady Koenig, **SM**, Atish Mitra, Abigail Stein*, Burt Todd. Unsupervised Topological Characterization of Slug, Churn and Annular Flow in Small-Diameter Vertical Pipes Using Euler Characteristic Surfaces, April 2026. Available at: arXiv:2604.06167 [cs.LG]
   • Under review at *International Journal of Multiphase Flow* (IJMF)

3. Kazuhiro Kawamura, **SM**, Atish Mitra. The Shadow of Vietoris–Rips Complexes in Limits, 29 pages, January 2026. Available at: arXiv:2601.01359 [math.AT]
   • Under review at *Foundations of Computational Mathematics* (FoCM)

2. Halley Fritze, **SM**, Marissa Masden, Atish Mitra, Michael Stickney. Faithful Reeb Graph Reconstruction of a Tectonic Subduction Zone from Earthquake Hypocenters, October 2024. Available at: arXiv:2410.19410 [cs.CG]
   • Under review at *La Matematica*

1. Rafal Komendarczyk, **SM**, Will Tran. Topological Stability and Latschev-type Reconstruction Theorems for Spaces of Curvature Bounded Above, 27 pages, June 2024. Available at: arXiv:2406.04259 [math.AT]
   • Under review at *Discrete & Computational Geometry* (DCG)

Refereed Journal Publications

13. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. A Closed-Form Persistence-Landmark Pipeline for Certified Point-Cloud and Graph Classification. *Transactions on Machine Learning Research* (TMLR), 2026. OpenReview: openreview.net/forum?id=4kZxNlE5Ve. Also available at: arXiv:2605.02836 [cs.LG]

12. Buddha Nath Sharma, Vishal Mandal, Salam Rabindrajit Luwang, Md. Nurujjaman, Atish J. Mitra, and **SM** (corresponding author). Interpretable Classification of Time Series using Euler Characteristic Surfaces. *Scientific Reports* (Nature Portfolio), 2026. DOI: 10.1038/s41598-026-64177-5. Also available at: arXiv:2603.15079 [cs.LG]

11. Rafal Komendarczyk, **SM**, Atish Mitra. Vietoris–Rips Shadow for Euclidean Graph Reconstruction. *Journal of Applied and Computational Topology* (JACT), vol. 10, no. 3, September 2026. DOI: 10.1007/s41468-026-00242-2. Also available at: arXiv:2506.01603 [math.AT]

10. Buddha Nath Sharma, Anish Rai, SR Luwang, Md. Nurujjaman, **SM**. Causality Analysis of COVID-19 Induced Crashes in Stock and Commodity Markets: A Topological Perspective. *International Journal of Modern Physics C*, 37(11):2650022, 2026. DOI: 10.1142/S0129183126500221. Also available at: arXiv:2502.14431 [q-fin.ST]

9. Henry Adams, Florian Frick, **SM**, Nicholas McBride*. Hausdorff VS Gromov–Hausdorff Distances. *Discrete & Computational Geometry* (DCG), February 2025. DOI: 10.1007/s00454-025-00722-9. Also available at: arXiv:2309.16648 [math.MG]

8. Kundan Mukhia, Anish Rai, SR Luwang, Md Nurujjaman, **SM**, Chittaranjan Hens. Complex Network Analysis of Cryptocurrency Market During Crashes. *Physica A: Statistical Mechanics and its Applications*, November 2024. DOI: 10.1016/j.physa.2024.130095. Also available at: arXiv:2405.05642 [q-fin.ST]

7. Anish Rai, Buddha Nath Sharma, Salam Rabindrajit Luwang, Md. Nurujjaman, **SM**. Identifying Extreme Events in the Stock Market: A Topological Data Analysis. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, October 2024. DOI: 10.1063/5.0220424. Also available at: arXiv:2405.16052 [q-fin.ST]

* Undergraduate student at the time of research or submission

6. **SM**. Demystifying Latschev's theorem: Manifold Reconstruction from noisy data. *Discrete & Computational Geometry* (DCG), May 2024. DOI: 10.1007/s00454-024-00655-9. Also available at: arXiv:2305.17288 [math.AT]

5. **SM** and Carola Wenk. Distance Measures for geometric graphs. *Computational Geometry: Theory and Applications* (CGTA), March 2024. DOI: 10.1016/j.comgeo.2023.102056. Also available at: arXiv:2209.12869 [cs.CG]

4. **SM**. Vietoris–Rips complexes of metric spaces near a metric graph. *Journal of Applied and Computational Topology* (JACT), May 2023. DOI: 10.1007/s41468-023-00122-z. Available at: arXiv:2204.14234 [math.AT]

3. **SM**, Jeffrey Vitter, and Carola Wenk. Approximating Gromov-Hausdorff distance in Euclidean space. *Computational Geometry: Theory and Applications* (CGTA), January 2024. DOI: 10.1016/j.comgeo.2023.102034. Also available at: arXiv:1912.13008 [math.MG]

2. Anish Rai, Ajit Mahata, Md Nurujjaman, **SM**, and Kanish Debnath. A sentiment-based modeling and analysis of stock price during the COVID-19: U- and Swoosh-shaped recovery. *Physica A: Statistical Mechanics and its Applications*. April 2022. DOI: 10.1016/j.physa.2021.126810. Also available at: arXiv:2110.03986 [q-fin.ST]

1. Brittany Terese Fasy, Rafal Komendarczyk, **SM**, and Carola Wenk. On the reconstruction of geodesic subspaces of $\mathbb{R}^N$. *International Journal of Computational Geometry & Applications*. September 2022. DOI: 10.1142/S0218195922500066. Also available at: arXiv:1810.10144 [math.AT]

Refereed Conference Proceedings

4. Henry Adams, **SM**, Fedor Manin, Nicolò Zava, Žiga Virk. Lower Bounding The Gromov–Hausdorff Distance In Metric Graphs. In Proceedings of the *42nd International Symposium on Computational Geometry (SoCG)*, LIPIcs vol. 367, Article 3, 2026. DOI: 10.4230/LIPIcs.SoCG.2026.3. Also available at: arXiv:2411.09182 [math.MG]

3. **SM**. Demystifying Latschev's theorem: Manifold Reconstruction from noisy data. In Proceedings of the *40th International Symposium on Computational Geometry (SoCG)*, Athens, Greece, June 11–14, 2024. Available at: https://doi.org/10.4230/LIPIcs.SoCG.2024.73

2. Erin Chambers, Brittany Fasy, Benjamin Holmgren*, **SM**, and Carola Wenk. Metric and path-connectedness properties of the Fréchet distance for paths and graphs. In Proceedings of the *35th Canadian Conference on Computational Geometry (CCCG)*, 2023. DBLP: conf/cccg/ChambersHFMW23. Available at: arXiv:2308.00900 [cs.CG]

1. **SM**. Graph mover's distance: An efficiently computable distance measure for geometric graphs. In Proceedings of the *35th Canadian Conference on Computational Geometry (CCCG)*, 2023. DBLP: conf/cccg/Majhi23. Available at: arXiv:2306.02133 [cs.CG]

Workshop Contributions

8. Pramita Bagchi, **SM**, Atish Mitra, Žiga Virk. Statistical Analysis of Persistence Landmarks: A Hilbert Space Embedding of the Space of Persistence Diagrams, *Fall Workshop on Computational Geometry*, Queens College, 2025.

7. Joshua Dorrington, **SM**, Atish Mitra, James Moukheiber*, Demi Qin, Jacob Sriraman*, Kristian Strommen. Topology of the Polar Vortex and Montana Weather. At *ATMCS'25*. June 2025. Available at: arXiv:2503.20743 [math.DS]

* Undergraduate student at the time of research or submission

6. Enrique Alvarado, Daniela Beckelhymer, Joshua Dorrington, Tung Lam, **SM**, Jasmine Noory, María Sánchez Muniz, Kristian Strommen. Detecting the Indian Monsoon using Topological Data Analysis. At *ATMCS'25*. June 2025. Available at: arXiv:2504.01022 [physics.ao-ph]

5. Halley Fritze, **SM**, Marissa Masden, Atish Mitra. Hausdorff–Approximation of Metric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, Tufts University, 2024. Available at: arXiv:2410.19410 [cs.CG]

4. E. Chambers, B. Fasy, B. Holmgren*, **SM**, and C. Wenk. Path-Connectivity of Fréchet Spaces of Graphs. Computational Geometry: Young Researchers Forum, 2022

3. **SM** and Carola Wenk. Distance Measures for Geometric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, 2022

2. Brittany Terese Fasy, **SM** and Carola Wenk. Threshold-based graph reconstruction using discrete Morse theory. In *Fall Workshop on Computational Geometry (FWCG)*, 2018. Abstract available at: Link

1. Brittany Terese Fasy, Rafal Komendarczyk, **SM**, and Carola Wenk. Topological reconstruction of metric graphs in $\mathbb{R}^N$. At *Fall Workshop on Computational Geometry (FWCG)*, 2017. Abstract available at: Link

Graduate Teaching Experience

- **Introduction to Data Science (DATS 6101)** **Fall 2026–*current***
  Data Science Program, George Washington University, USA
  Topics: exploratory data analysis, probability, expectation and variance, sampling distributions, confidence intervals, and hypothesis testing. *Course notes:* smajhi.com/DATS-6101

- **Linear Algebra for Data Science** **Summer 2024**
  Data Science Program, George Washington University, USA
  Course notes: smajhi.com/lin-alg
  Topics: linear algebra foundations, Gaussian elimination, pivoting techniques for stability, LU decomposition, eigenvalues and eigenvectors, solving linear systems using iterative methods, projections, linear regression, and the SVD.

- **Algorithm Design (DATS 6001)** **Spring 2024–*current***
  Data Science Program, George Washington University, USA
  Course notes: smajhi.com/DATS-6001
  Topics: basics of computational complexity, data structures (array, stack, queue, tree, etc), algorithms (search, sort, etc), and programming paradigms like dynamic programming and greedy algorithms.

- **Topological Data Analysis (CS)** **Fall 2023**
  National Institute of Technology, Sikkim, India
  Topics: topological spaces, metric spaces and their examples, simplicial complexes, homology in $\mathbb{Z}/2$ coeff, persistent homology, bottleneck and Wasserstein distance, and non-linear time-series analysis using TDA. *Course notes:* smajhi.com/TDA

- **Computer Science Foundations (DATS 6450)** **Fall 2023**
  Data Science Program, George Washington University, USA
  Topics: computer design, programming in Python, object-oriented programming

- **Introduction to Data Mining (DATS 6103)** **Fall 2023–*current***
  Data Science Program, George Washington University, USA
  Topics: data wrangling, linear and logistic regression, classification, clustering, data visualization, support vector machines, machine learning algorithms. *Course notes:* smajhi.com/DATS-6103

* Undergraduate student at the time of research or submission

- **Statistics for Data Science (DATASCI 203)** **August 2020–July 2023**
  School of Information, University of California, Berkeley, USA
  Topics: probability theory, sampling distributions, estimators and convergence theorems, confidence intervals, testing, and regression.

- **Linear Algebra, Complex Analysis** **January 2013–June 2013**
  Christ University and Scimetric Pvt Ltd, Bangalore, India

- **Analysis, Linear Algebra, Complex Analysis** **November 2011–July 2012**
  GATE-IIT Coaching Institute, JP Nagar, Bangalore, India
  Graduate level, for competitive national exams, e.g., National Eligibility Test

- **Analysis and Linear Algebra** **February 2012–July 2012**
  MES College, Department of Mathematics, Malleswaram, Bangalore, India

Undergraduate Teaching Experience

- **Data Science for All** **Summer 2025**
  Data Science Program, George Washington University, USA
  Topics: Foundations of data science for undergraduate students

- **Data Science Capstone (DATS 4001)** **Fall 2024–*current***
  Data Science Program, George Washington University, USA
  Goal: The goal of the capstone is for the students to apply the knowledge acquired throughout the GWU Data Science undergraduate program to solve a real-world problem, using real data. It typically starts with developing S.M.A.R.T. questions and goes through the entire data science life cycle in multiple iterations.

- **Undergraduate Statistics for Business Students (MATH 1140)** **Summer 2019**
  Tulane University, USA
  Topics: sampling methods, descriptive statistics, probability theory, random variables, limit theorems, confidence intervals, hypothesis testing, and linear regression.

Other Teaching

- **Graduate Teaching Assistant** **Fall 2014–Spring 2017**
  Tulane University, USA

Invited Talks and Accepted Abstracts

- **Title:** The Mixup Euler Characteristic Profile: A Functorial Topological Descriptor via Euler Calculus, Stability, and Integral Geometry **November 16–20, 2026**
  Invited minisymposium presentation, SIAM Conference on Mathematics of Data Science (MDS26), Salt Palace Convention Center, Salt Lake City, UT

- **Title:** Beyond a Shadow of a Doubt: Certified Graph Reconstruction in $\mathbb{R}^N$ **October 3, 2026**
  AMS Special Session on Advances in Applied Topology, AMS Fall Eastern Sectional Meeting, George Washington University, Washington, D.C.

- **Title:** PALACE: Closed-Form, Isometry-Invariant Certificates for Geometric Classification via Adaptive-Landmark Persistence Kernels **July 9, 2026**
  Workshop on Rigidity Theory meets Geometric Data Science, ICERM, Brown University, Providence, RI

- **Title:** Lower Bounding the Gromov–Hausdorff Distance in Metric Graphs **June 5, 2026**
  42nd International Symposium on Computational Geometry (SoCG), Rutgers University, New Brunswick, NJ

- **Title:** PLACE: Persistence Landmarks for Assured Classification Embedding **May 20, 2026**
  Workshop on Foundations of Computational Geometry and Topology, ICERM, Brown University (video)

- **Title:** Geometric Latschev's Theorem **March 11, 2026**
  Spring Topology Conference, University of Alabama
- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction **January 7, 2026**
  Joint Mathematics Meetings, Washington, D.C.
- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction **Nov 8, 2025**
  Fall Workshop on Computational Geometry, Queens College
- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction **Oct 4, 2025**
  AMS Southeastern Sectional Meeting, Tulane University
- **Title:** Topological and Geometric Reconstruction of Spaces Beyond Manifolds **June 19, 2025**
  TDA Week, Kyoto University, Japan, 2025
- **Title:** Topological Stability and Latschev-type Reconstruction Theorems for CAT(κ) Spaces **March 8, 2025**
  AMS Southeastern Sectional Meeting, Clemson University
- **Title:** Topological Stability and Latschev-type Reconstruction Theorems for Spaces of Curvature Bounded Above **March 6, 2025**
  Spring Topology and Dynamics, Christopher Newport University, Newport News
- **Title:** Predicting the Onset and Withdrawal of the Indian Monsoon using Persistent Homology **January 9, 2025**
  AMS Special Session: Climate Science at the Interface Between Topological Data Analysis and Dynamical Systems Theory, Joint Mathematics Meetings, Seattle, 2025
- **Title:** Lower Bounding the Gromov–Hausdorff distance in Metric Graphs **Nov 14–15, 2024**
  The 31st Fall Workshop on Computational Geometry, Tufts University
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes **Oct 14, 2024**
  Indian Institute of Technology, Mandi, India
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes **Oct 8, 2024**
  Vellore Institute of Technology, Chennai, India
- **Title:** Demystifying Latschev's Theorem **June 14, 2024**
  Symposium on Computational Geometry, Athens, Greece
- **Title:** Demystifying Latschev's Theorem **March 28, 2024**
  Montana State University
- **Title:** Demystifying Latschev's Theorem **August 23, 2023**
  Applied Algebraic Topology Research Network (AATRN) (video)
- **Title:** Graph Mover's Distance **August 2–4, 2023**
  The 35th Canadian Conference on Computational Geometry, Montreal, Canada
- **Title:** Similarity Measures for Geometric Graphs **October 14–15, 2022**
  The 30th Fall Workshop on Computational Geometry
  North Carolina State University, USA
- **Title:** Topological Methods in the Reconstruction and Comparison of Shapes **February, 2022**
  Mathematics Department, ICFAI University, Tripura, India
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes **September, 2021**
  Department of Mathematics, Hunter College, NY, USA
- *Fall Workshop on Computational Geometry, Queens College, New York, USA* **October, 2018**
- *Fall Workshop on Computational Geometry, SUNY (Stony Brook), New York, USA* **November, 2017**

PhD Dissertation Committee

- Peiqi Yang, Mathematics, George Washington University **Summer 2025**
- Will Tran, Mathematics, Tulane University; thesis: *Distortion and Curvature in the Shape Reconstruction Problem* **Spring 2024**

PhD Advisees

- Buddha Nath Sharma, National Institute of Technology Sikkim, India; co-advised with Md. Nurujjaman; thesis: *Applications of topological data analysis in financial markets* **August 2025–*current***

Graduate Student Mentees

- Alexander D. Silberman (MS data science, GW): `akriti`—statistical inference for topological data analysis **August 2026–*current***
- Sayam Palrecha (MS data science, GW): topological data analysis for finance; now Data Science Research Specialist, GW Regulatory Studies Center **2025–2026**
- Mauricio Montes (PhD, Auburn University): statistical analysis of Euler characteristic surfaces **March–August 2025**
- Khush Shah and Shikha Kumari (MS data science, GW): matching geometric graphs with graph neural networks; now Junior Data Scientist at Cintra and AI/ML & Software Engineer at Nexas 8 International, respectively **August 2023–December 2024**
- Anish Rai (PhD, National Institute of Technology Sikkim, India): predicting stock market crashes with topological data analysis; now postdoctoral scholar, Chennai Mathematical Institute **August 2022–August 2024**

Undergraduate Student Mentees

- Edward Bae (GW): `akriti`—statistical inference for topological data analysis **August 2026–*current***
- Joe Opitz and Adam Moser (Montana Tech; with Atish Mitra): topological data analysis for finance **2026–*current***
- Abby Stein (GW); now master's student in applied mathematics, Johns Hopkins University **August 2025–2026**
- Jacob Sriraman (Montana Technological University); now PhD student, Montana State University **August 2025–2026**
- James Moukheiber (GW); now master's student, University of Zurich **August 2024–2026**

Department/University Service

- Chair, Undergraduate Curriculum Committee
  George Washington University, Data Science Program **August 2025–*current***
- Undergraduate Advisor
  George Washington University, Data Science Program **August 2024–*current***

Awards, Scholarships, Recognition

- **Travel Grant**, University of California, Berkeley
- **UGC-CSIR NET Research Fellowship**, India, June 2012
- **TIFR Junior Research Fellowship** for Integrated PhD studies at TIFR-CAM, Bangalore. August 2009.
- **Secured grade "A" in SCIENCE TALENT SEARCH EXAMINATION**
  conducted by JATIYA VIJNAN PARISAD and INDIAN SCIENCE CONGRESS ASSOCIATION

Academic Services

- With Atish Mitra and Tom Needham, I am organizing the Applied Topology and TDA track of the 60th Spring Topology and Dynamics Conference, Ohio State, March 2027.
- I organized two special sessions on "Topological Data Analysis for Non-linear Dynamics" and "Topological and Geometric Shape Reconstruction" at the Joint Mathematics Meetings, Washington DC, January 2026.

- I co-organized a special session on "Advances in Applied Topology and Topological Data Analysis" at the AMS southeastern sectional meeting at Tulane University in 2025.
- I have been a reviewer for journals, like Foundations of Computational Mathematics, Discrete and Computational Geometry, SIAM Journal on Mathematics of Data Science, Journal of Applied and Computational Topology, Foundations of Data Science, Journal of Combinatorial Optimization, and International Journal of Computational Geometry.
- I have been a reviewer for conferences, like the **International Conference on Learning Representations (ICLR)**, 2027, the **International Conference on Artificial Intelligence and Statistics (AISTATS)**, 2027, the International Symposium on Computational Geometry, Algebraic Topology: Methods, Computation, and Science (ATMCS), International Symposium on Spatial and Temporal Databases, European Workshop on Computational Geometry, ACM International Conference on Advances in Geographic Information Systems, WADS Algorithms and Data Structures Symposium.
- I organized SIAM Graduate Student Chapters at Tulane University.
- I organized a data science webinar series at the University of California, Berkeley.

Computational Skills

Java, C, R, Python, Ruby, JavaScript, SQL, Bash.

Software Projects

- `akriti` akriti.io | GitHub
  Founder and lead. An open-source Python ecosystem supplying the statistical inference layer the TDA stack lacks—hypothesis tests, effect sizes, per-region significance, sample sizes—on top of GUDHI, Ripser and Hera. Apache-2.0, under written governance, with student developers at GW.
- **Simplicial Complexes in JS** GitHub
  Simplicial-complex computations in JavaScript, including homology via Smith normal form.
- **Shape Reconstruction Visualization** WebApp | GitHub
  My PhD reconstruction algorithm for planar metric graphs, as a web app (JavaScript, HTML, CSS).

Popular Media

- Interviewed by GWU Hatchet for an Article **August 2024**

Entrepreneurial Experience

- **Scimetric Edulabs Private Limited** (Bangalore), **co-founder** and **director** **December 2012–April 2017**
  A start-up that coached science students for national PhD and academic entrance tests, through franchises with several private colleges; six trainers.

Hobby

I enjoy playing the piano and classical guitar.